



Final-Year Project

Pricing and Calibration of VIX Options under a Forward Variance Curve Model

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Abstract

This project investigates the pricing and calibration of European VIX options within a forward-variance framework inspired by Bergomi's work. The objective is twofold: first, to simulate VIX option prices using Monte Carlo methods; second, to calibrate a mixture-of-exponentials model to market-observed implied-volatility smiles. The final implementation handles both calls and puts, compares the single-factor Bergomi model with a mixture-of-exponentials extension, and performs both single-maturity and multi-maturity calibration. The results show that the single-factor model generates an excessively flat smile, whereas the mixture-of-exponentials specification captures the positive skew observed in VIX options significantly better. After calibration, the mean squared error becomes very small for the maturity considered, while the multi-maturity extension confirms the model's ability to track the term structure of the VIX smile.

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1 Introduction

Volatility plays a central role in derivatives pricing. It directly affects option prices, but it can also be viewed as a financial underlying in its own right when considering VIX-linked products. The VIX measures the market's expectation of S&P 500 volatility over an approximately thirty-day horizon. VIX options are therefore derivatives whose underlying is not the price level of a conventional asset, but a measure of future volatility.

This project is set within the framework of forward-variance models. The key idea is to model the dynamics of the forward-variance curve directly rather than relying solely on the dynamics of the underlying asset. This approach belongs to the literature on dynamic smile models, whose objective is not only to reproduce a volatility surface at a given date, but also to generate coherent implied-volatility dynamics [4]. It is consistent with Bergomi's framework, which emphasises treating forward variances and volatility instruments as market objects in their own right [1, 3]. The approach is particularly well suited to the VIX, since the index can be interpreted as the square root of an average forward variance over a thirty-day window.

The analysis proceeds in several stages. First, market data are cleaned in order to extract implied-volatility smiles for VIX calls and puts. A single-factor Bergomi model is then implemented. The model relies on an exponentially decaying volatility-of-variance function. It is simple, Markovian and computationally efficient, but produces a very flat smile, which is a well-known limitation for VIX options. To address this issue, the implementation introduces a mixture-of-exponentials model inspired by [2], allowing a positive slope closer to that observed in the market. Finally, the model is calibrated to market implied volatilities using least squares and subsequently extended to a multi-maturity calibration.

The main contributions of the final implementation are therefore:

- joint treatment of calls and puts;
- an explicit comparison between market data, the single-factor Bergomi model and the mixture-of-exponentials model;
- calibration of the mixture model for a given maturity;
- an extension to sequential calibration across several maturities;
- summary tables and validation charts.

2 Theoretical Framework

2.1 VIX Options

The VIX is an index representing the market's expectation of volatility over a future period of approximately thirty days. Unlike conventional options written directly on an asset price, VIX options are derivatives whose underlying is a measure of future volatility. This definition is consistent with the practical construction of the VIX, which is based on S&P 500 options and documented in market references [12, 11].

In this project, we focus on vanilla VIX options. For maturity T and strike K , the payoff of a VIX call is:

$$(VIX_T - K)^+.$$

The payoff of a VIX put is:

$$(K - VIX_T)^+.$$

Under the risk-neutral measure, the time- t price of a VIX call with maturity T is:

$$C_t = \mathbb{E} \left[e^{-r(T-t)} (VIX_T - K)^+ \mid \mathcal{F}_t \right].$$

Throughout the remainder of the project, interest rates are assumed to be zero, i.e. $r = 0$. The price therefore becomes:

$$C_t = \mathbb{E} \left[(VIX_T - K)^+ \mid \mathcal{F}_t \right].$$

Puts are treated analogously. To compute implied volatilities, the implementation uses Black's formula applied to VIX futures, consistent with the standard convention for options on futures [6, 2].

2.2 Forward Variance

We denote by ξ_t^T the instantaneous forward variance observed at time t for future horizon T . This quantity represents the time- t expectation of the underlying asset's future instantaneous variance at time T . Direct modelling of forward variance lies at the core of Bergomi's work on smile dynamics [1, 3].

The forward variance over the interval $[T_1, T_2]$, observed at time T , is defined as:

$$V_T^{T_1, T_2} = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \xi_T^u du.$$

For the VIX, we consider a future window of length δ , corresponding to approximately thirty days. Hence:

$$VIX_T^2 = \frac{1}{\delta} \int_T^{T+\delta} \xi_T^u du, \quad \delta = \frac{1}{12}.$$

Thus, the VIX can be interpreted as the square root of an average future forward variance.

More generally, consider an option on forward variance with payoff:

$$\Psi \left(V_T^{T_1, T_2} \right),$$

où :

$$V_T^{T_1, T_2} = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \xi_T^u du.$$

Its time- t price is given by:

$$Price_t = \mathbb{E} \left[e^{-r(T-t)} \Psi \left(V_T^{T_1, T_2} \right) \mid \mathcal{F}_t \right].$$

Since we assume $r = 0$, this becomes:

$$Price_t = \mathbb{E} \left[\Psi \left(V_T^{T_1, T_2} \right) \mid \mathcal{F}_t \right].$$

2.3 Bergomi Model

The Bergomi model describes the stochastic dynamics of instantaneous forward variance. For any future horizon $T > 0$, the process $(\xi_t^T)_{0 \leq t \leq T}$ satisfies:

$$d\xi_t^T = \xi_t^T \varphi(T-t) dW_t, \quad 0 \leq t \leq T,$$

with:

$$\xi_0^T = \xi_0^{T, mkt}.$$

Here, $(W_t)_{t \geq 0}$ is a standard Brownian motion and φ is a deterministic function satisfying appropriate integrability conditions.

Explicit Solution of the Equation

Applying Itô's formula to $\log(\xi_t^T)$, and since:

$$d\xi_t^T = \xi_t^T \varphi(T-t)dW_t,$$

on a :

$$\frac{d\xi_t^T}{\xi_t^T} = \varphi(T-t)dW_t.$$

By Itô's formula:

$$d\log(\xi_t^T) = \frac{1}{\xi_t^T} d\xi_t^T - \frac{1}{2(\xi_t^T)^2} d\langle \xi^T \rangle_t.$$

Or :

$$d\langle \xi^T \rangle_t = (\xi_t^T)^2 \varphi^2(T-t)dt.$$

Therefore:

$$d\log(\xi_t^T) = \varphi(T-t)dW_t - \frac{1}{2}\varphi^2(T-t)dt.$$

Integrating from 0 to t yields:

$$\log(\xi_t^T) = \log(\xi_0^T) + \int_0^t \varphi(T-s)dW_s - \frac{1}{2} \int_0^t \varphi^2(T-s)ds.$$

Hence:

$$\xi_t^T = \xi_0^T \exp \left(\int_0^t \varphi(T-s)dW_s - \frac{1}{2} \int_0^t \varphi^2(T-s)ds \right).$$

This expression guarantees the positivity of forward variance.

Reduction of the Pricing Problem

On suppose :

$$0 \leq t \leq T \leq T_1 < T_2.$$

From the explicit solution, for any $u \geq T$, we have:

$$\xi_T^u = \xi_0^u \exp \left(\int_0^T \varphi(u-s)dW_s - \frac{1}{2} \int_0^T \varphi^2(u-s)ds \right).$$

We split the integrals over $[0, t]$ and $[t, T]$:

$$\int_0^T \varphi(u-s)dW_s = \int_0^t \varphi(u-s)dW_s + \int_t^T \varphi(u-s)dW_s,$$

and:

$$\int_0^T \varphi^2(u-s)ds = \int_0^t \varphi^2(u-s)ds + \int_t^T \varphi^2(u-s)ds.$$

Therefore:

$$\xi_T^u = \xi_t^u \exp \left(\int_t^T \varphi(u-s)dW_s - \frac{1}{2} \int_t^T \varphi^2(u-s)ds \right).$$

Define:

$$Z_{t,T}^u = \exp \left(\int_t^T \varphi(u-s)dW_s - \frac{1}{2} \int_t^T \varphi^2(u-s)ds \right).$$

Hence:

$$\xi_T^u = \xi_t^u Z_{t,T}^u.$$

The variable ξ_t^u is $\mathcal{F}_t$ -measurable, since it depends only on information available up to time t . The variable $Z_{t,T}^u$ depends solely on Brownian increments over $[t, T]$. By independence of Brownian increments, $Z_{t,T}^u$ is independent of $\mathcal{F}_t$.

We therefore have:

$$\mathbb{E} \left[\Psi \left(V_T^{T_1, T_2} \right) \mid \mathcal{F}_t \right] = \mathbb{E} \left[\Psi \left(\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \xi_T^u du \right) \mid \mathcal{F}_t \right].$$

Using $\xi_T^u = \xi_t^u Z_{t,T}^u$, we obtain:

$$\mathbb{E} \left[\Psi \left(V_T^{T_1, T_2} \right) \mid \mathcal{F}_t \right] = \mathbb{E} \left[\Psi \left(\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \xi_t^u Z_{t,T}^u du \right) \mid \mathcal{F}_t \right].$$

Since the curve $(\xi_t^u)_{u \in [T_1, T_2]}$ is known at time t , it may be treated as deterministic conditional on $\mathcal{F}_t$. We therefore define the functional:

$$\Psi(t, x(\cdot)) = \mathbb{E} \left[\Psi \left(\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} x(u) \exp \left(\int_t^T \varphi(u-s) dW_s - \frac{1}{2} \int_t^T \varphi^2(u-s) ds \right) du \right) \right].$$

We therefore obtain:

$$\mathbb{E} \left[\Psi \left(V_T^{T_1, T_2} \right) \mid \mathcal{F}_t \right] = \Psi \left(t, (\xi_t^u)_{u \in [T_1, T_2]} \right).$$

This derivation shows that pricing an option on forward variance reduces to evaluating an expectation involving only future Brownian increments.

Exponential Case and Markovian Representation

In the single-factor Bergomi model, we choose:

$$\varphi(\tau) = \omega e^{-k\tau}, \quad \omega > 0, \quad k > 0.$$

We aim to show that:

$$\int_0^\tau \varphi(u-s) dW_s = \varphi(u-\tau) X_\tau,$$

where X is an Ornstein–Uhlenbeck process:

$$dX_t = -kX_t dt + dW_t, \quad t \geq 0, \quad X_0 = 0.$$

The explicit solution to this equation is:

$$X_\tau = \int_0^\tau e^{-k(\tau-s)} dW_s.$$

Then:

$$\varphi(u-\tau) X_\tau = \omega e^{-k(u-\tau)} \int_0^\tau e^{-k(\tau-s)} dW_s.$$

Combining the exponential terms:

$$\varphi(u-\tau) X_\tau = \omega \int_0^\tau e^{-k(u-\tau)} e^{-k(\tau-s)} dW_s.$$

Or :

$$e^{-k(u-\tau)} e^{-k(\tau-s)} = e^{-k(u-s)}.$$

Therefore:

$$\varphi(u-\tau) X_\tau = \omega \int_0^\tau e^{-k(u-s)} dW_s.$$

Since:

$$\varphi(u - s) = \omega e^{-k(u-s)},$$

we obtain:

$$\varphi(u - \tau)X_\tau = \int_0^\tau \varphi(u - s)dW_s.$$

Hence:

$$\int_0^\tau \varphi(u - s)dW_s = \varphi(u - \tau)X_\tau.$$

This representation is computationally important because it replaces direct simulation of the stochastic integral with simulation of a Gaussian random variable associated with the Ornstein–Uhlenbeck process.

2.4 Mixture-of-Exponentials Model

The single-factor Bergomi model provides a coherent framework for simulating forward-variance dynamics, but it generally produces an implied-volatility smile that is too flat relative to VIX market observations. This limitation motivates the extensions proposed in Bergomi’s work on the volatility-of-volatility smile [2, 5].

To obtain more flexible dynamics, we introduce a mixture-of-exponentials model. Define:

$$G \left[\int_0^t \varphi(T - s)dW_s \right] = \exp \left(\int_0^t \varphi(T - s)dW_s - \frac{1}{2} \int_0^t \varphi^2(T - s)ds \right).$$

The mixture-of-exponentials model is then given by:

$$\xi_t^T = \xi_0^T \left[(1 - \gamma)G \left[\int_0^t \omega_1 e^{-k(T-s)}dW_s \right] + \gamma G \left[\int_0^t \omega_2 e^{-k(T-s)}dW_s \right] \right],$$

with:

$$\omega_1 > 0, \quad \omega_2 > 0, \quad k > 0, \quad \gamma \in (0,1).$$

The parameters ω_1 and ω_2 represent two different levels of volatility of variance. The parameter γ controls the relative weight of each component in the mixture.

In practice, when ω_1 and ω_2 are sufficiently far apart, the model can generate a more pronounced positive skew, which is characteristic of the implied-volatility smiles observed in VIX options.

3 Numerical Methodology

This section describes in greater detail how the mathematical model is translated into the final Python implementation. The numerical objective is to construct a usable VIX implied-volatility curve from market quotes and then generate a model curve that can be compared directly with the market. Comparing implied volatilities rather than prices provides a more intuitive quoting convention for options while remaining consistent with standard VIX futures-option market practice [6, 2].

3.1 Data Import and Preparation

The data are obtained from Cboe delayed quotes. For each maturity, the dataset contains strikes, bid and ask prices, trading volumes and quoted implied volatilities. As in standard procedures used to construct VIX-type volatility measures, option quotes must first be made suitable for analysis: representative prices are computed, illiquid quotes are removed, and relevant strikes around the reference future are retained [11, 12].

The reference value of an option is its mid price:

$$Mid = \frac{Bid + Ask}{2}.$$

This avoids working directly with either the bid or the ask, which would respectively represent a lower and an upper bound for the transaction price. The mid price therefore provides a more neutral approximation of the market price.

In the implementation, the data-cleaning process consists of the following steps:

- removal of unusable header rows from the downloaded file;
- conversion of numerical columns to the appropriate format;
- separation of calls and puts;
- retention of options with strictly positive trading volume;
- removal of zero, missing or anomalous implied volatilities;
- restriction to a liquid and calibration-relevant strike range;
- aggregation of duplicate observations by strike to obtain a single market implied volatility for each strike.

This step is important because nonlinear calibration is highly sensitive to input quality. A very illiquid option with a wide bid–ask spread or an inconsistent implied volatility can distort the objective function and degrade the calibrated parameters. In line with standard smile-calibration practice, data preparation is therefore an integral part of the numerical modelling process rather than merely a technical preprocessing step [3, 10].

3.2 Construction of the Market Smile

Once the data have been cleaned, the implementation constructs call and put smiles separately. This separation provides a consistency check. For the same maturity and strike range, implied volatilities derived from calls and puts should be close when market quotes are internally consistent. In the results obtained, the two curves are almost superimposed, supporting the use of the extracted smile as the calibration target.

The market VIX future is estimated from option quotes and subsequently used as the underlying in Black’s formula. In the charts, it is represented by a vertical line around:

$$F_0^{VIX} \approx 20.87.$$

This line separates strikes close to at-the-money from more distant strikes. From a financial perspective, strikes above the futures level correspond to scenarios in which the market prices a substantial increase in future volatility. These regions contain a significant share of the information about the stress premium embedded in VIX options.

3.3 Simulation of the Single-Factor Bergomi Model

For the single-factor Bergomi model, the implementation exploits the Markovian representation of the Ornstein–Uhlenbeck factor. Instead of discretising the entire stochastic integral, the Gaussian random variable is simulated directly:

$$X_T \sim \mathcal{N}\left(0, \frac{1 - e^{-2kT}}{2k}\right).$$

For each Monte Carlo path and each grid point u over the interval $[T, T + \delta]$, forward variance is computed as:

$$\xi_T^u = \xi_0^u \exp \left(\omega e^{-k(u-T)} X_T - \frac{1}{2} \omega^2 e^{-2k(u-T)} \frac{1 - e^{-2kT}}{2k} \right).$$

This formulation guarantees the positivity of forward variance. It is directly related to Bergomi's forward-variance models, in which the variance curve is treated as a market object with its own dynamics [1, 3].

The simulated VIX is then obtained by numerically integrating forward variance over thirty days:

$$VIX_T^2 \approx \frac{1}{\delta} \sum_{j=1}^n \xi_T^{u_j} \Delta u, \quad u_j \in [T, T + \delta].$$

On en déduit :

$$VIX_T = \sqrt{VIX_T^2}.$$

In the implementation, this operation is repeated over a large number of paths, producing a simulated sample of possible VIX values at maturity.

3.4 Simulation of the Mixture-of-Exponentials Model

The mixture-of-exponentials model follows the same simulation logic but replaces the single lognormal dynamics with a combination of two components:

$$\xi_T^u = \xi_0^u [(1 - \gamma)G_1(u, T) + \gamma G_2(u, T)].$$

The two factors G_1 and G_2 are constructed using two different volatility-of-variance levels, ω_1 and ω_2 . Numerically, this gives the model a more asymmetric terminal VIX distribution than the single-factor specification. This asymmetry is essential for generating an upward-sloping smile, since high-strike options are particularly sensitive to extreme volatility-spike scenarios.

This choice is consistent with the approach proposed in [2], where mixtures of exponentials introduce additional flexibility while preserving a numerically tractable Markovian structure. This feature is central to the final implementation: it explains why, even before calibration, the mixture model produces a curve substantially closer to the market than the single-factor model.

3.5 Monte Carlo Pricing of Calls and Puts

Using the simulated values $VIX_T^{(i)}$, call and put prices are estimated by empirical averaging. For a call with strike K :

$$\hat{C}_0(K, T) = \frac{1}{N} \sum_{i=1}^N (VIX_T^{(i)} - K)^+.$$

For a put:

$$\hat{P}_0(K, T) = \frac{1}{N} \sum_{i=1}^N (K - VIX_T^{(i)})^+.$$

Since interest rates are assumed to be zero in the project, no discount factor is applied. This assumption simplifies the analysis and remains reasonable for the short maturities considered.

The model-implied VIX future is estimated simultaneously as:

$$\hat{F}_0^{VIX} = \frac{1}{N} \sum_{i=1}^N VIX_T^{(i)}.$$

The model-implied future is then compared with the market future. This diagnostic is useful because a model may reproduce the slope of the smile while still generating an incorrect forward level. In the results, the mixture model produces a future closer to the market level than the single-factor model, representing an improvement even before calibration.

3.6 From Prices to Implied Volatilities

Monte Carlo prices are not compared directly with market prices in the charts. They are first converted into implied volatilities by inverting Black's formula, the standard convention for options on futures [6]. For a call:

$$C_{Black} = F_0^{VIX} N(d_1) - KN(d_2),$$

with:

$$d_1 = \frac{\ln(F_0^{VIX}/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

For a put:

$$P_{Black} = KN(-d_2) - F_0^{VIX} N(-d_1).$$

The inversion consists in finding the value of σ for which the Black price equals the Monte Carlo price. The implementation uses a robust numerical root-finding procedure, with boundary checks and handling of cases in which inversion fails. This step is required to display the model smile in the same units as the market smile.

3.7 Definition of the Objective Function

Calibration is formulated as a least-squares problem on implied volatilities. For a given maturity, the objective function is:

$$MSE(\theta) = \frac{1}{m} \sum_{j=1}^m (IV_{\text{mod}}(K_j; \theta) - IV_{\text{mkt}}(K_j))^2,$$

where m is the number of retained strikes and:

$$\theta = (\omega_1, \omega_2, k, \gamma).$$

A squared-error criterion is standard in smile-calibration problems. It penalises large deviations and provides a simple measure of overall goodness of fit. Similar methods are used in many calibrations of parametric volatility models [10, 8, 9].

3.8 Parameter Constraints and Transformations

The model parameters must satisfy the following natural constraints:

$$\omega_1 > 0, \quad \omega_2 > 0, \quad k > 0, \quad \gamma \in (0.1).$$

To prevent the optimiser from leaving the admissible domain, the implementation optimises transformed parameters:

$$\omega_1 = e^{a_1}, \quad \omega_2 = e^{a_2}, \quad k = e^{a_3},$$

and:

$$\gamma = \frac{1}{1 + e^{-a_4}}.$$

The constraints are therefore satisfied automatically. This technique improves calibration stability because the optimiser works with unconstrained variables while always producing financially admissible parameter values.

3.9 Single-Maturity Calibration

The first calibration is performed for a single maturity. The objective is to verify that the mixture-of-exponentials model can accurately reproduce an observed VIX smile. Starting from an initial parameter set, the implementation simulates prices for each strike, converts them into implied volatilities, computes the error relative to the market, and adjusts the parameters until the MSE is minimised.

This step also helps interpret the resulting parameters. In particular, the ratio ω_1/ω_2 measures the separation between the two volatility-of-variance regimes. A high ratio indicates that the model requires an extreme component to reproduce high-strike option prices.

3.10 Multi-Maturity Calibration

The final step generalises the procedure across several maturities. The implementation selects sufficiently liquid maturities, retains only those with an adequate number of valid strikes, and applies the mixture-model calibration to each maturity.

The objective is no longer simply to fit an isolated curve, but to assess model stability across a small VIX volatility surface. The results are summarised in a table containing, for each maturity, the time to maturity T , calibrated parameters, model-implied future, market future and MSE. This presentation makes it possible to assess both fit quality and parameter evolution across maturities.

In the final implementation, the parameter k is fixed or strongly stabilised in the multi-maturity calibration. This prevents excessive instability from one maturity to another. It effectively preserves a common time scale for exponential decay, while ω_1 , ω_2 and γ absorb most of the differences in smile shape across maturities.

4 Numerical Results

4.1 Market Smile: Calls and Puts

The first figure shows the market implied-volatility smile for calls and puts. The two curves are very close across the strike range considered. This provides an initial consistency check: the selected market data do not exhibit any major inconsistency between the call and put sides of the option chain.

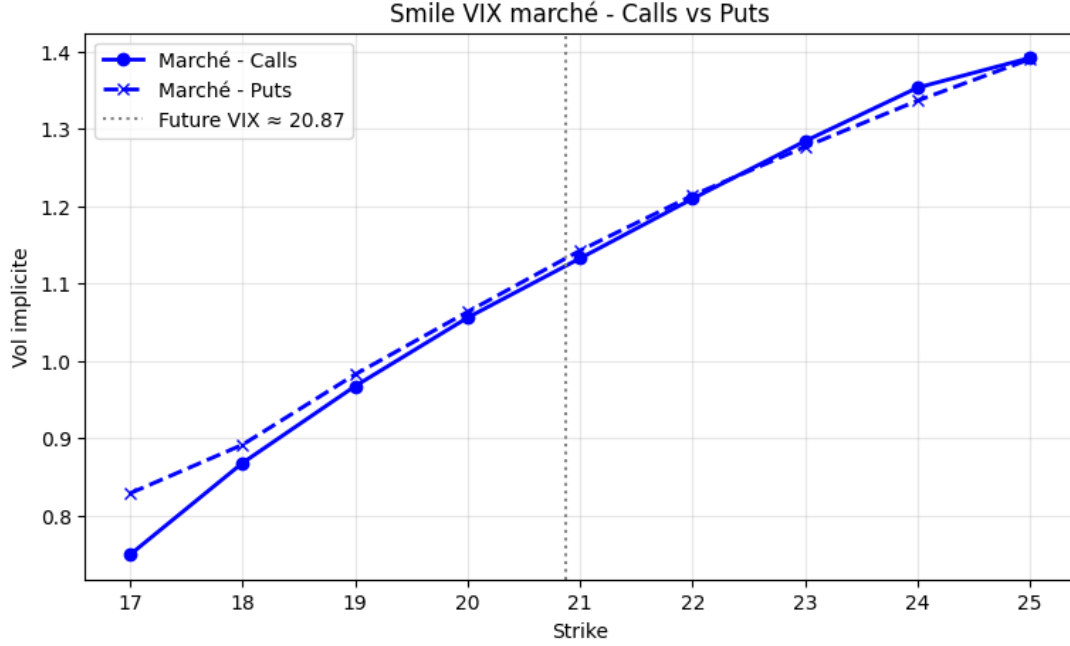


Figure 1: Smile VIX de market : comparison calls and puts.

The market VIX future is approximately:

$$F_0^{VIX} \approx 20.87.$$

The vertical line around this level separates the near-the-money region from more distant strikes. Implied volatility is already high at low strikes, around 0.75–0.85, and then rises almost steadily to approximately 1.40 for the highest strikes. Financially, this positive slope means that the market assigns a high premium to options that gain value during sharp increases in the VIX. In other words, investors place significant value on protection against episodes of market stress.

The near overlap between call- and put-implied volatilities is also important for the subsequent analysis. It indicates that the model can be calibrated to a representative smile without a material contradiction between the two option types. In practice, the remaining small differences may arise from bid–ask spreads, liquidity differences or quote noise.

4.2 Bergomi Model single-factor

The single-factor Bergomi model is then compared with the market smile. The result is clear: the model produces an almost horizontal curve at approximately 62% implied volatility, whereas the market displays a strongly upward-sloping smile.

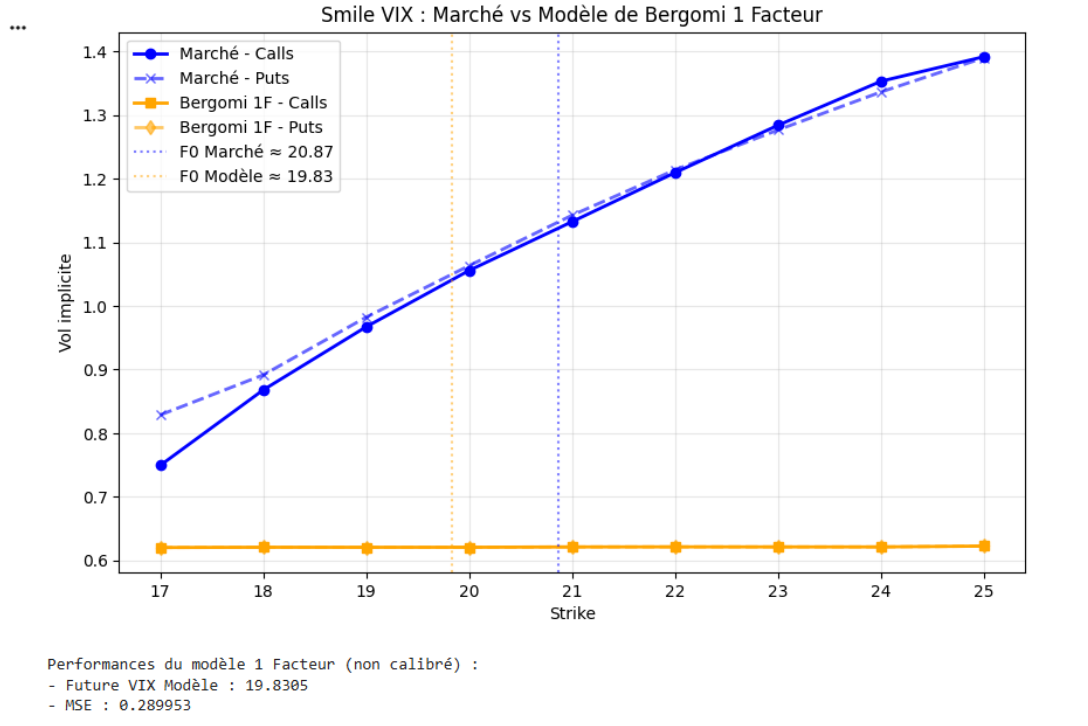


Figure 2: VIX smile: market vs. single-factor Bergomi model.

The numerical results for the uncalibrated model are:

$$F_0^{VIX,model} \approx 19.83, \quad MSE_{1F} = 0.289953.$$

The model-implied future is below the market future, but the main discrepancy is not merely the forward level. The principal issue is the shape of the smile. The single-factor model substantially underestimates market implied volatilities, with the underestimation becoming even more pronounced at high strikes.

From a financial perspective, this means that the single-factor model assigns insufficient probability, or insufficient pricing weight, to extreme VIX-spike scenarios. It generates a terminal distribution that is too regular and too close to a simple lognormal behaviour. VIX options, however, embed a significant premium for abrupt volatility moves. This result illustrates the theoretical limitation highlighted by Bergomi: an overly rigid forward-variance structure cannot adequately control the volatility-of-volatility smile [1, 2, 3].

4.3 Uncalibrated Mixture-of-Exponentials Model

The mixture-of-exponentials model substantially improves the fit. Even before calibration, the model curve exhibits a positive slope and lies much closer to the market smile.

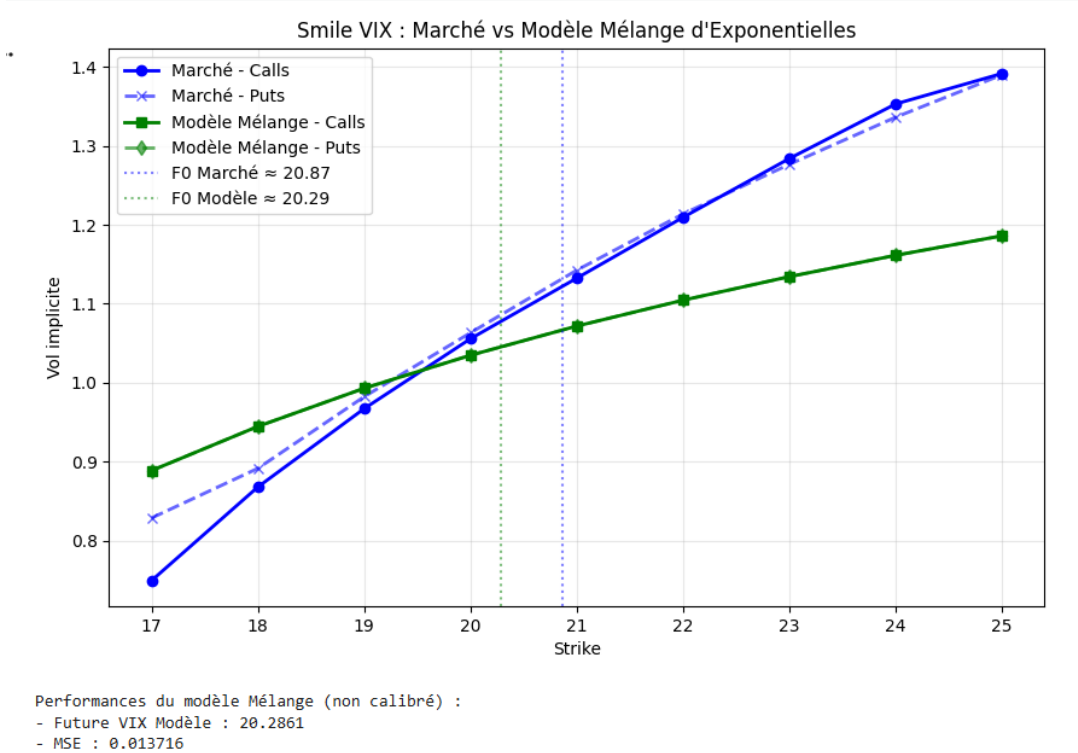


Figure 3: VIX smile: market vs. mixture-of-exponentials model before calibration.

The uncalibrated results are:

$$F_0^{VIX,model} \approx 20.29, \quad MSE_{mix} = 0.013716.$$

The improvement in MSE is substantial:

$$0.289953 \longrightarrow 0.013716.$$

The model-implied future is also closer to the market future than under the single-factor model.

Graphically, the mixture model follows the general market trend well between strikes 17 and 22, but remains too low for the highest strikes. Thus, although the mixture introduces more realistic asymmetry, the initial parameters still assign insufficient weight to very high-volatility scenarios. This naturally motivates calibration: the model structure is appropriate, but its parameters must be adjusted to the data.

4.4 Overall Model Comparison

The following figure overlays the market smile, the single-factor Bergomi model and the mixture-of-exponentials model, clearly illustrating the contribution of the model extension.

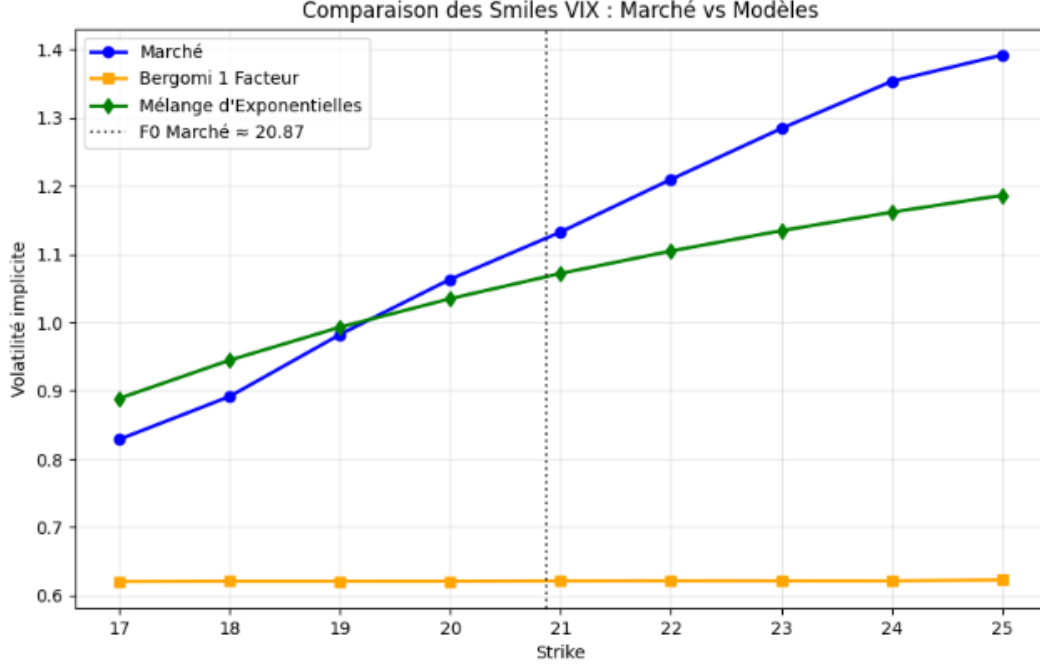


Figure 4: Overall comparison of VIX smiles: market, single-factor Bergomi and mixture-of-exponentials models.

Table 1: Comparison of implied volatilities before calibration.

Strike	Market	Bergomi single-factor	Mélange d'exponentielles
17	82.88%	62.04%	88.88%
18	89.16%	62.08%	94.47%
19	98.24%	62.08%	99.30%
20	106.33%	62.08%	103.48%
21	113.25%	62.13%	107.18%
22	120.95%	62.15%	110.46%
23	128.40%	62.14%	113.44%
24	135.31%	62.14%	116.14%
25	139.16%	62.27%	118.61%

The table highlights two effects. First, the single-factor model is almost insensitive to strike, remaining around 62% across the entire range. Second, the mixture model can generate a positive slope, but the slope remains insufficient in the right wing of the smile. For example, at strike 25, the market implied volatility is 139.16%, compared with 118.61% for the uncalibrated mixture model.

This difference is financially significant. High strikes represent extreme-volatility scenarios. If the model undervalues this region, it will underprice far out-of-the-money VIX calls, precisely the instruments used to hedge against market shocks. Accurate calibration of the right wing is therefore essential for both pricing and risk management.

4.5 Single-Maturity Calibration

The single-maturity calibration adjusts the mixture-of-exponentials parameters to minimise the discrepancy with the market smile. The resulting optimised parameters are:

$$\omega_1 = 9.132852, \quad \omega_2 = 1.014084, \quad k = 0.377582, \quad \gamma = 0.727190.$$

The ratio between the two volatility-of-variance parameters is:

$$\frac{\omega_1}{\omega_2} = 9.006.$$

RÉSULTATS DE CALIBRATION
 omega1 optimisé = 9.132852
 omega2 optimisé = 1.014084
 k optimisé = 0.377582
 gamma optimisé = 0.727190
 Ratio w1/w2 = 9.006

Figure 5: Optimised parameters from the single-maturity calibration.

This ratio is informative. It shows that the calibration requires two markedly different regimes: a moderate component associated with ω_2 and a much more volatile component associated with ω_1 . The high-volatility component raises the right wing of the smile and incorporates the pricing of extreme scenarios. The parameter $\gamma \approx 0.727$ indicates that the mixture assigns substantial weight to the second component while retaining enough of the first component to generate the required convexity.

After calibration, the final error becomes:

$$MSE_{finale} = 0.000094.$$

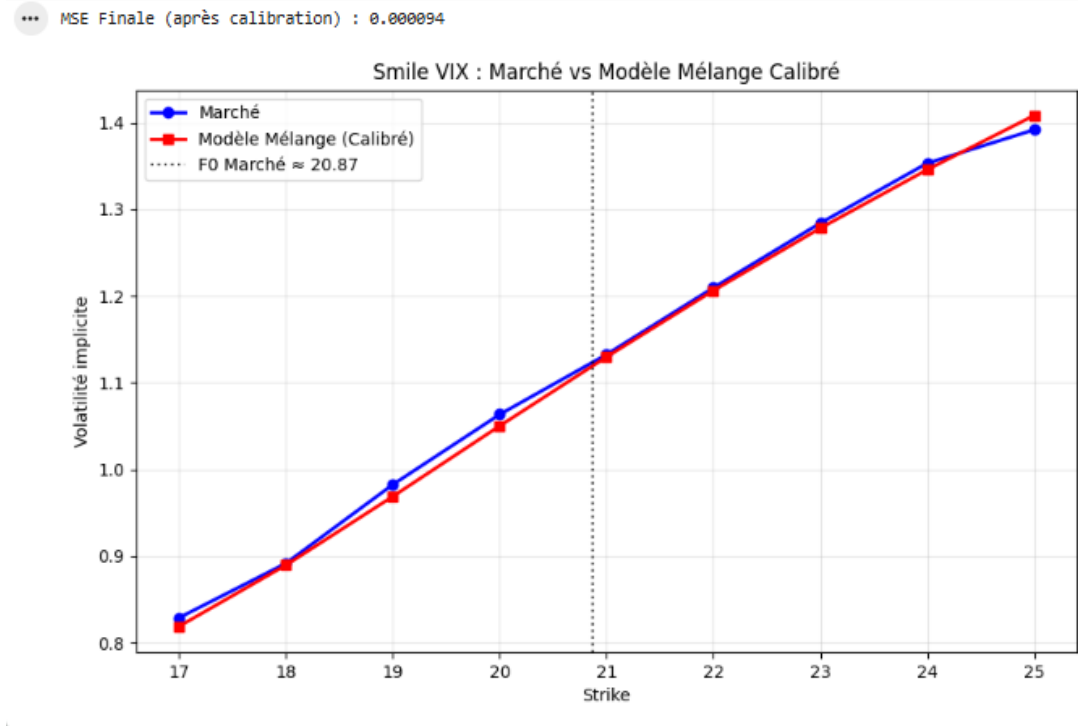


Figure 6: VIX smile: market vs. calibrated mixture-of-exponentials model.

The chart shows an almost complete overlap between the market and the calibrated model. The model reproduces both the overall level and the slope of the smile. The fit is particularly

strong around the futures level, i.e. near strike 21, and remains highly satisfactory in the wings. The small discrepancy at the highest strike is limited relative to the overall amplitude of the smile.

Financially, this result indicates that the mixture-of-exponentials model can reconstruct an implied VIX distribution consistent with market prices. Calibration does not merely correct an average level; it adjusts the entire shape of the smile, which is essential for a usable pricing model.

4.6 Multi-Maturity Calibration

The final part of the implementation extends the calibration across several maturities. The selected maturities are:

Wed Apr 15 2026, Wed Apr 22 2026, Wed Apr 29 2026.

The implementation retains maturities with a sufficient number of valid strikes:

Wed Apr 15 2026 : 20 valid strikes, $F_0^{mkt} = 20.46$,
Wed Apr 22 2026 : 15 valid strikes, $F_0^{mkt} = 21.06$,
Wed Apr 29 2026 : 14 valid strikes, $F_0^{mkt} = 20.86$.

```
*** Wed Apr 15 2026 retenue : 20 strikes valides (F0 Marché = 20.46)
    Wed Apr 22 2026 retenue : 15 strikes valides (F0 Marché = 21.06)
    Wed Apr 29 2026 retenue : 14 strikes valides (F0 Marché = 20.86)
```

Figure 7: Selected maturities and numbers of valid strikes.

Table 2: Summary of the multi-maturity calibration.

Maturity	T	ω_1	ω_2	k	γ	F_0^{mod}	MSE
Wed Apr 15 2026	0.0055	18.5153	1.4674	0.3776	0.8799	20.4600	0.000573
Wed Apr 22 2026	0.0247	11.7759	1.4385	0.3776	0.8156	21.0600	0.000538
Wed Apr 29 2026	0.0438	9.0102	0.9936	0.3776	0.7239	20.8575	0.000136

The following figure shows the fit quality across the three maturities.

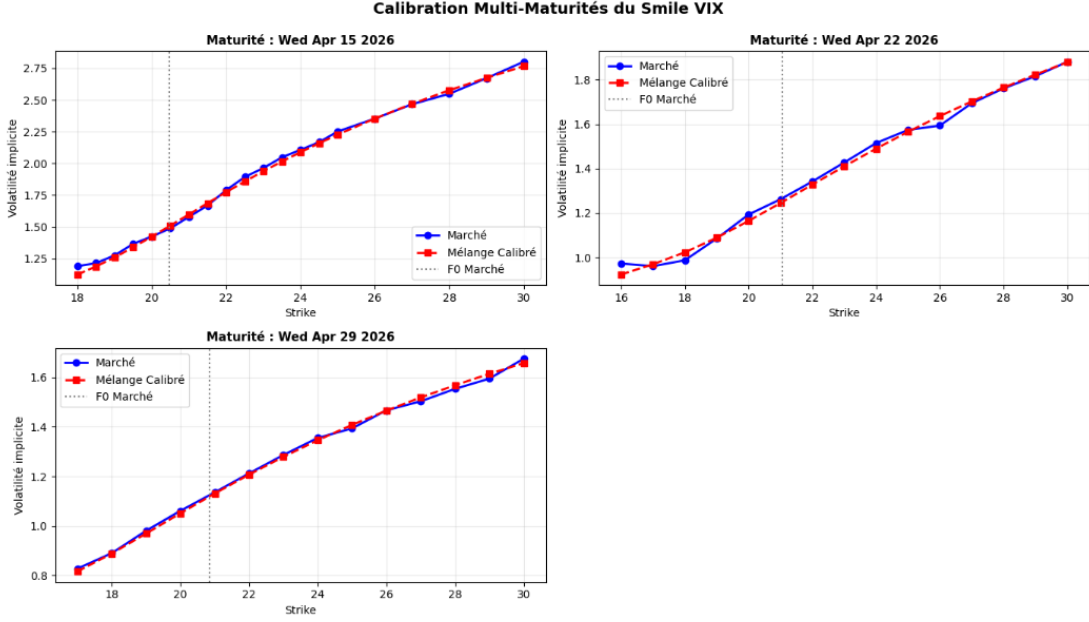


Figure 8: Multi-Maturity Calibration du smile VIX.

The three charts show that the calibrated model tracks the market smiles very closely. Fit quality is particularly strong for the 29 April 2026 maturity, where the MSE falls to 0.000136. For the 15 and 22 April maturities, the error is slightly higher but remains very small. The remaining discrepancies are concentrated at a few strikes and may arise from market noise, liquidity effects or Monte Carlo error.

The evolution of the parameters is also noteworthy. The parameter ω_1 decreases with maturity:

$$18.5153 \longrightarrow 11.7759 \longrightarrow 9.0102.$$

The weight γ also decreases:

$$0.8799 \longrightarrow 0.8156 \longrightarrow 0.7239.$$

This evolution has a natural financial interpretation. Shorter maturities require a stronger volatility-of-variance component because short-dated smiles are often more sensitive to immediate volatility shocks. As maturity increases, the impact of extreme scenarios is spread over a longer horizon, reducing the need for an explosive component.

Keeping k constant at approximately 0.3776 provides a common decay time scale across the three calibrations. This stabilises the resulting surface and prevents each maturity from being fitted using entirely different dynamics. In practice, such a constraint is useful because a perfect maturity-by-maturity fit can produce unstable parameters that are difficult to interpret.

Overall, the multi-maturity calibration confirms that the mixture-of-exponentials model is not limited to fitting a single curve. It can also track a small term structure of VIX smiles, strengthening its relevance for volatility-option pricing.

5 Discussion

5.1 Financial Interpretation

The results confirm an important point: an overly simple forward-variance model is insufficient to reproduce VIX smiles. The single-factor Bergomi model generates coherent dynamics but imposes too rigid a structure on the variance curve. In practice, VIX option implied volatilities

embed a substantial premium for stress scenarios, reflected in the pronounced positive slope of the smile.

The mixture of exponentials introduces two volatility scales. The ω_1 component captures rapid and violent variance moves, while ω_2 represents more moderate dynamics. This separation is useful for representing the coexistence of normal and extreme scenarios in VIX option prices. This is precisely the appeal of Bergomi forward-variance models: they model the dynamics of variance instruments directly rather than deriving the entire volatility structure from a single spot-volatility factor [1, 2, 3].

The steep smile observed in the data therefore indicates that high-strike VIX calls are not merely far out-of-the-money options; they represent insurance against episodes of market stress. Investors purchasing such options often seek protection against a sharp rise in S&P 500 implied volatility. The fact that the single-factor model substantially underestimates this region of the smile shows that it specifically undervalues this stress insurance.

5.2 Link to Recent Financial-Market Developments

The project is directly related to recent financial-market developments, as the VIX remains one of the most closely watched measures of uncertainty in US equity markets. In spring 2025, tariff announcements by the Trump administration triggered a period of elevated volatility across global markets. Reuters described rapid equity selling, a shift towards safe-haven assets and large market moves associated with US tariff announcements [13]. The Federal Reserve Bank of St. Louis also analysed the episode as a temporary but pronounced increase in financial-market volatility, noting that the VIX represents a thirty-day implied volatility derived from S&P 500 option prices [14].

This example illustrates the relevance of the project. During a political or macroeconomic shock, the level of the VIX alone may not be sufficient; the shape of the VIX option smile also matters. A rise in the VIX indicates that the market expects higher average volatility, while a very steep smile additionally indicates that investors are paying a substantial premium for extreme scenarios. The model is designed to reproduce this market information through the parameters of the mixture-of-exponentials specification.

Developments in 2026 also show that the VIX can remain relatively low despite a complex macroeconomic environment. In late May 2026, the VIX returned to its lowest level in several months amid supportive equity-market conditions [15]. At the same time, Federal Reserve officials continued to note that geopolitical and energy shocks could alter the outlook for US monetary policy [16]. This coexistence of apparent calm in spot VIX and persistent macroeconomic uncertainty reinforces the value of studying VIX options: their smiles contain more granular information than the index level alone.

5.3 Calibration Quality

The single-maturity calibration produces a final MSE of 0.000094, indicating a very close fit between the model and the market. The multi-maturity calibration also produces low errors, all below 6×10^{-4} in the results obtained.

Nevertheless, the results should be interpreted with caution. Calibration is performed on a limited number of maturities and strikes. The resulting parameters therefore depend on data cleaning, the selected strike range, the number of Monte Carlo simulations and the numerical stability of Black implied-volatility inversion. As in any inverse calibration problem, different parameter sets may sometimes generate similar smiles, so the final error must be considered alongside parameter stability and interpretability [8, 9, 10].

5.4 Model Limitations

Several limitations should be noted:

- the model assumes zero interest rates;
- the initial forward-variance curve is simplified;
- calibration is performed on implied volatilities rather than directly on bid–ask prices;
- Monte Carlo noise may affect the objective function;
- the parameter k is sometimes fixed or strongly constrained in the multi-maturity calibration;
- the model does not explicitly capture dependence between the VIX and the S&P 500.

These limitations do not invalidate the results, but they point to natural directions for improvement. They also explain why a good visual fit should not be interpreted as complete model validation. A market model should be assessed not only by its ability to reproduce observed prices, but also by the stability of its parameters and the consistency of the dynamics it generates for hedging instruments [3].

5.5 Potential Improvements

Several extensions could be considered:

- increase the number of Monte Carlo paths to reduce statistical noise;
- use antithetic variates or other variance-reduction techniques;
- calibrate to mid prices with liquidity or bid–ask-spread weighting;
- introduce a non-constant initial forward-variance curve;
- calibrate several maturities simultaneously with parameters smoothed across maturity;
- compare the model with an SVI or SSVI approach for implied-volatility smiles;
- reconstruct a local-volatility surface from the implied surface using Dupire’s formula, in order to compare a local-volatility approach with the forward-variance framework [7, 10].

6 Conclusion

This project investigated the pricing and calibration of VIX options under a forward variance curve model. The first step consisted in constructing the market smile from option quotes. Calls and puts exhibit very similar implied volatilities, supporting the consistency of the selected data and allowing these observations to be used as calibration targets.

The single-factor Bergomi model was then implemented. Its Markovian representation through an Ornstein–Uhlenbeck process enables efficient Monte Carlo simulation and a simple representation of forward variance. However, the results show that the model produces a smile that is far too flat. Its mean squared error before calibration is high:

$$MSE_{1F} = 0.289953.$$

This limitation is financially important because it means that the model undervalues high-strike options, i.e. the instruments that provide protection against sharp increases in volatility.

The mixture-of-exponentials model substantially improves the results. Even before calibration, its error falls to:

$$MSE_{\text{mix}} = 0.013716.$$

After calibration, the final error becomes:

$$MSE_{\text{finale}} = 0.000094.$$

The calibrated model reproduces the market smile very accurately in both level and slope. The parameter interpretation is consistent with the observed shape: the model requires a high volatility-of-variance component to generate the right wing of the smile and reflect the pricing of stress scenarios.

Finally, the multi-maturity calibration supports the robustness of the approach. The three selected maturities are fitted well, with low errors and coherent parameter evolution. Short maturities require a higher volatility-of-variance component, consistent with the idea that short-dated smiles react more strongly to immediate market shocks.

Overall, the project shows that mixture-of-exponentials forward-variance models provide a relevant framework for VIX option pricing. They connect an observable market object—the implied-volatility smile—with interpretable forward-variance dynamics. For production use, the model would require greater numerical precision, a broader market dataset and additional stabilisation of its numerical outputs.

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